

For my final project, I decided to attempt at formalizing a theorem from abstract algebra in Lean4. The theorem formally states, “Let R and S be rings, and let $f: R \rightarrow S$ be a bijective ring homomorphism. Then R and S are ring isomorphic.”

I ended up choosing this theorem because I had just learned it around the same time we started dwelling on proof formalizations in this class. Not only that, after some searching, I saw that there was no real formalization of this theorem, but just a constructor that was used to skip the whole proof process. Because this theorem was something that I already understood and knew how to prove, I finalized this as my project topic. To prove this theorem, we show that the inverse function of f preserves ring operations and is therefore also a ring homomorphism. Once the inverse function is shown to be a ring homomorphism, the original function and its inverse together form a ring isomorphism.

After defining the inverse function, I proved that it behaves as an actual inverse of f . The first lemma establishes the right inverse property, showing that applying f to the chosen preimage returns the original element of S . The second lemma establishes the left inverse property, relying on the injectivity of f .

The next portion of the project was proving that the inverse function preserves the ring operations. To do this, I separated the proof steps into four different lemmas. First, I showed that the inverse maps zero to zero. Next, I showed that it maps one to one. Then, I proved that it preserves addition and multiplication. These proofs use the injectivity of f together with the fact that f is already known to be a ring homomorphism.

After proving these lemmas, I used them to define the inverse function as a ring homomorphism. I showed that the inverse function preserved all ring operations and therefore

was a ring homomorphism. The last step was to put everything together and define a ring isomorphism. I used the original homomorphism, the inverse function, and the proofs to do this.

All in all, I would say the part that took the most amount of time for me was actually deciding on what theorem to formalize. It felt like every time I decided on a theorem, it had already been formalized in Lean. For the coding portion, because we were given access to useful resources such as GitHub Copilot, the Lean community Zulip, and the professor's own ongoing formalization project, the process was a lot easier than I had anticipated. Although the theorem was not particularly difficult, having to break it down and prove all the smaller bits made me realize just how complicated a proof really is. Overall, this project helped me appreciate simplified proofs and also gain a better understanding of Lean.